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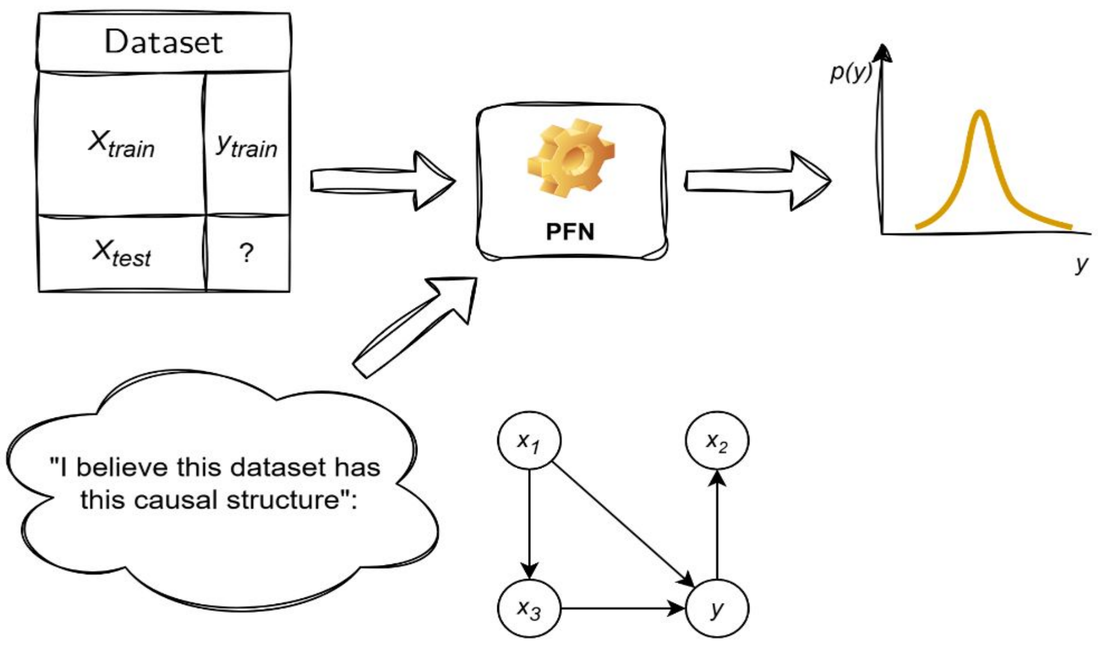
Bayesian Tabular Few-shot Learning with Causal Information

Ole Ossen[1], Jake Robertson[1,2], Arik Reuter[3,4], Magnus Bühler[1], Lennart Purucker[1,2], Frank Hutter[1,2,5]

[1]University of Freiburg [2]Prior Labs [3]University of Cambridge [4]Max Planck Institute for Intelligent Systems [5]ELLIS Institute Tübingen

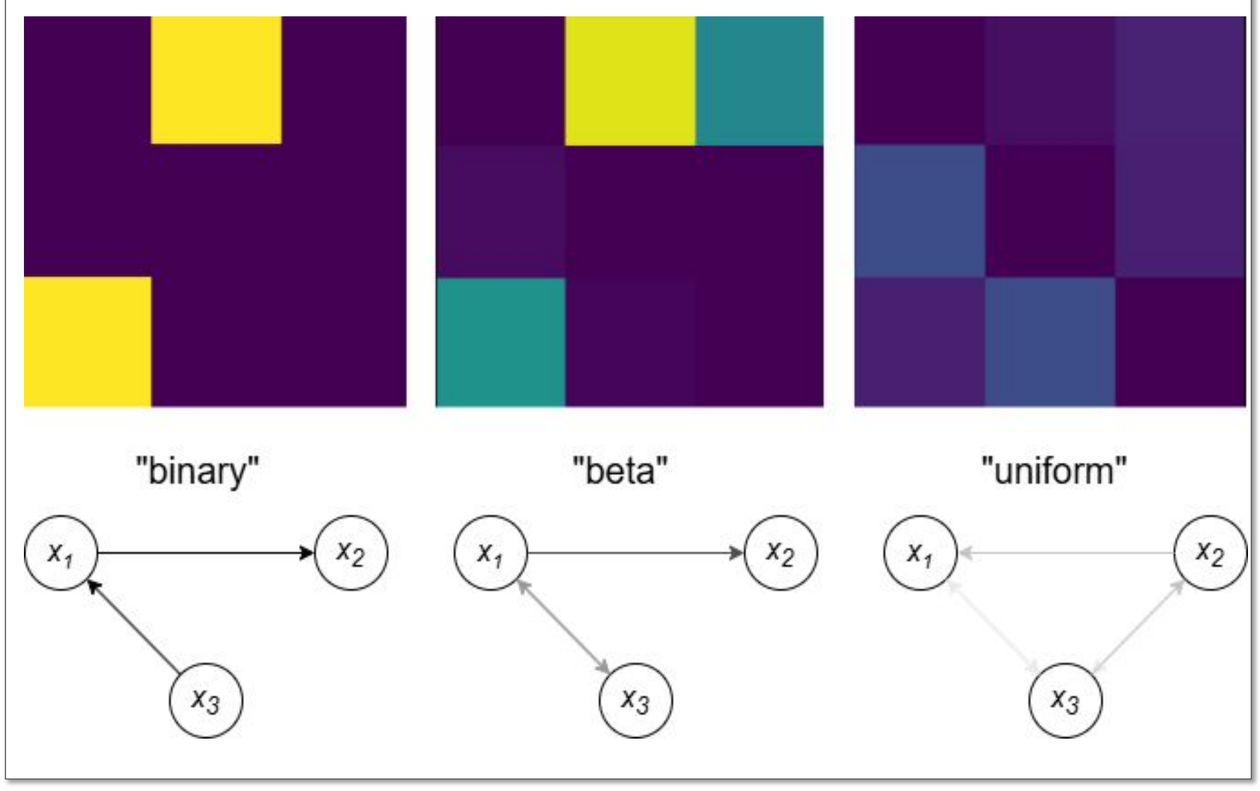
Idea

To make tabular few-shot prediction easier for foundation models, we give them a hint: our prior belief about the causal structure of the dataset.



Graph information

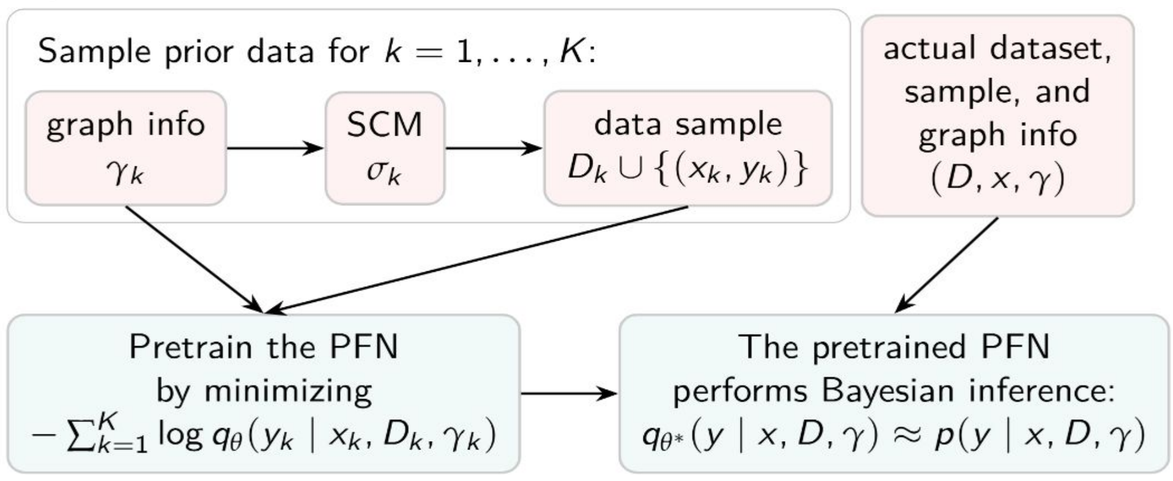
The model is given a **probabilistic adjacency matrix**. We can use different uncertainty levels:



Methodology

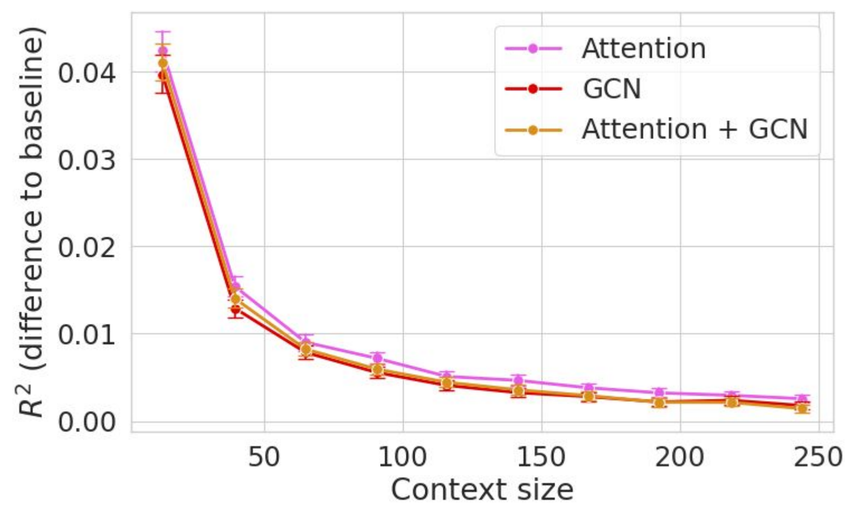
The training follows the usual PFN paradigm. Graph info is processed using either of the following:

- **Causally restricted attention.** The model's feature-wise attention is restricted to edges present in the causal graph.
- **GCN encoding.** The graph is encoded using a graph convolutional network whose output modulates the model's layer norm layers.

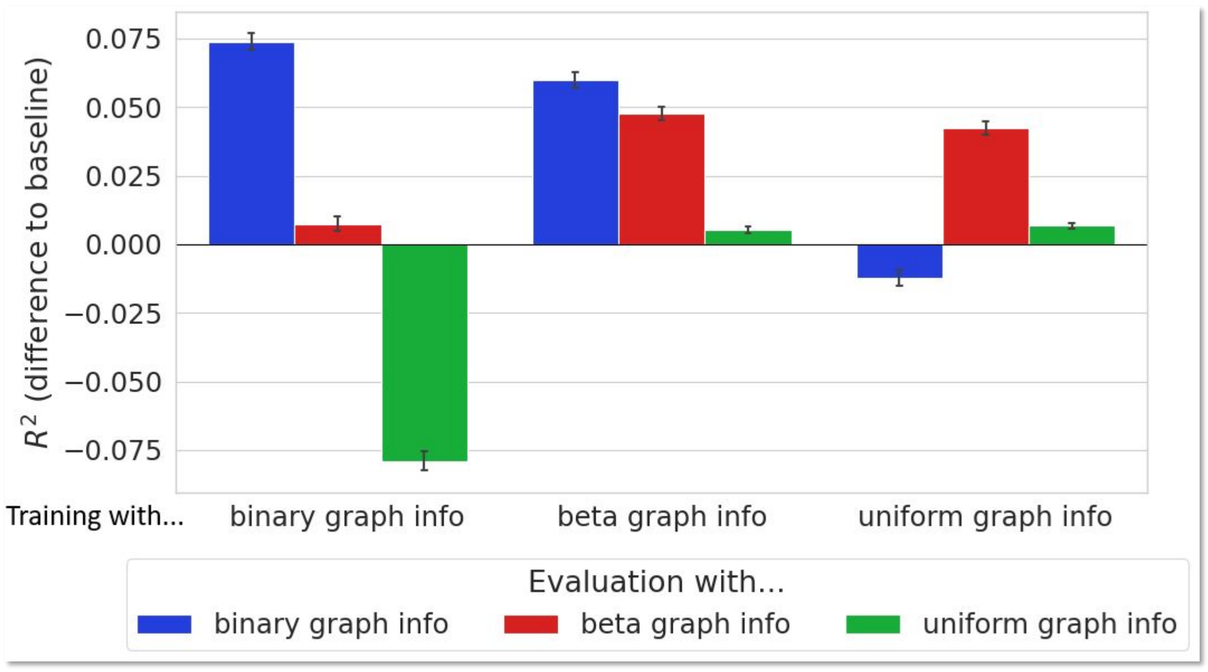


Bayesian interpretation

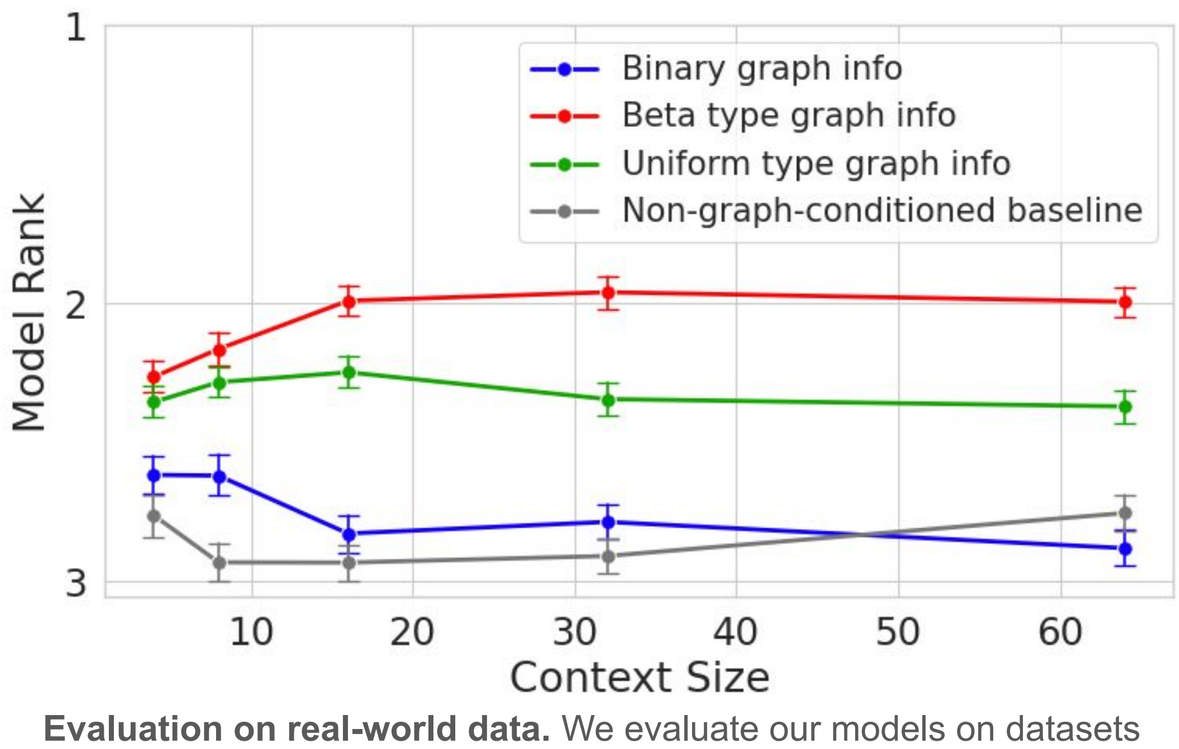
Our graph-conditioning method is most useful in the few-shot setting. As the number of samples increases, the data overpowers the prior, and causal information doesn't help much.



Results



Evaluating and training models on different kinds of graph information. Our models trained with binary graph information perform the best, but only when they are given the correct, binary graph information during inference. The models trained on other types of graph information are more robust when provided other types.



Evaluation on real-world data. We evaluate our models on datasets subsampled from TabArena regression tasks. Our graph-conditioning models outperform the baseline. For each dataset, one fixed piece of graph information is obtained by running causal discovery methods on a 500 row subsample of the dataset.